Abstract

This study examines the extent to which measurable Instagram post characteristics predict engagement rate using quantitative modeling techniques. Drawing on a publicly available Instagram Analytics Dataset, the analysis evaluates the predictive value of caption length, hashtag count, posting day and time, media type, and content category. A structured methodology incorporating descriptive statistics, visual exploration, and multiple linear regression was applied to assess feature-level relationships. Results indicate that engagement rate exhibited minimal variance across all examined variables, with regression models demonstrating low explanatory power and non-significant coefficients. These findings led to a failure to reject the null hypothesis, suggesting that the selected post characteristics do not meaningfully predict engagement rate within this dataset. The study concludes that engagement outcomes are likely influenced by factors not captured in the available data, such as content quality, audience characteristics, creator influence, or platform-level algorithmic dynamics. Recommendations emphasize the need for future research incorporating richer predictors, audience-level metrics, and advanced modeling techniques to more effectively capture the complexity of engagement behavior on social media platforms.

Introduction

Instagram has become one of the most influential digital platforms for communication, branding, and audience engagement. With over a billion active users, the platform provides organizations with a powerful channel to distribute visual content and interact with consumers. Engagement metrics, including likes, comments, shares, and saves serve as indicators of audience interest and are central to how Instagram's algorithm determines content visibility. As machine learning and data analytics continue to evolve, researchers have increasingly focused on understanding the factors that influence engagement and how predictive models can support more effective content strategies. Recent studies demonstrate that machine learning algorithms can successfully forecast impressions and engagement by analyzing variables such as media type, posting time, and historical interaction patterns (Singh et al., 2025). These models help marketers optimize content by identifying which features most strongly influence reach and interaction.

Despite the availability of rich social media data, many organizations still struggle to translate raw metrics into actionable insights. Engagement patterns shift rapidly due to algorithmic updates, cultural trends, and user behavior changes.

Research in social media analytics emphasizes the importance of multidimensional feature analysis, including content attributes, timing, and contextual factors to accurately predict engagement outcomes (Madnure & Kadam, 2025). Without a structured analytical framework, organizations often rely on intuition or inconsistent historical patterns, resulting in inefficient content strategies. As Polonsky & Waller (2019) note, organizations frequently encounter challenges when they fail to clearly define the variables influencing a problem or neglect to apply systematic research methods.

This study addresses these challenges by developing a predictive model capable of estimating Instagram engagement rate using post-level features. Engagement rate, calculated as interactions normalized by reach, provides a more equitable basis for comparison across posts with varying audience sizes and aligns with best practices in digital marketing analytics. By analyzing historical Instagram post data, this research identifies which content characteristics most strongly predict engagement and how these variables interact to influence audience behavior. The findings will support organizations in shifting from reactive posting habits to proactive, data-driven content planning. Ultimately, this research contributes to a deeper understanding of engagement dynamics and offers a structured analytical approach that organizations can use to strengthen their digital communication strategies.

Objectives

The primary aim of this research is to identify the content characteristics that most strongly predict Instagram engagement and to determine how these variables interact to influence audience behavior. The project also seeks to develop a quantitative predictive modeling framework that uses machine learning techniques to estimate engagement rate based on measurable features such as caption length, number of hashtags, posting day and time, and media type. By examining these variables collectively, the study aims to determine the extent to which engagement outcomes can be forecasted using structured analytical methods.

A second objective is to evaluate the performance of multiple predictive models including regression and tree-based machine learning algorithms, to determine which approach provides the most accurate and interpretable results. This includes assessing model performance using metrics such as R^2, RMSE, and feature importance scores, as well as validating assumptions and ensuring optimal model performance.

A third objective is to generate actionable insights that organizations can use to optimize their content strategies. By identifying the features that most strongly influence engagement rate, the study aims to support evidence-based decision-making in digital marketing contexts, helping organizations improve content planning, strengthen audience engagement, and enhance overall communication effectiveness.

Together, these objectives support the broader purpose of the study: to develop a reliable, data-driven framework for predicting Instagram engagement and to contribute to a deeper understanding of the factors that shape audience interaction on social media platforms.

Overview

This study examines how specific Instagram post characteristics influence engagement rate and evaluates the extent to which these outcomes can be predicted using quantitative modeling techniques. The research uses a publicly available Instagram Analytics Dataset containing post-level metrics such as media type, caption length, hashtags, posting time, and audience size. These variables provide a multidimensional view of content performance and support both descriptive and predictive analysis. Engagement rate, calculated as interactions normalized by reach, serves as the primary outcome variable, offering a standardized measure that accounts for differences in audience size and visibility.

The study is grounded in the growing need for organizations to understand and forecast engagement in an increasingly competitive digital environment. Predictive modeling bridges the gap between creative decision-making and data-driven insight generation.

To address this need, the study applies a structured analytical approach that integrates statistical modeling and machine learning techniques. Multiple linear regression was used to evaluate the statistical significance of post-level features, while tree-based models such as decision trees and random forests will be used to capture nonlinear relationships and interactions. Model performance will be assessed using metrics such as R^2, RMSE, and feature importance scores to ensure both accuracy and interpretability. A rigorous data preparation process including cleaning, feature engineering, and variable transformation, ensures that the dataset is analytically sound and aligned with model assumptions.

Overall, this study provides a systematic framework for understanding the factors that shape Instagram engagement and for developing predictive models that can support evidence-based content strategy. By integrating descriptive analysis, statistical modeling, and machine learning, the research aims to generate actionable insights that organizations can use to optimize their social media communication efforts.

Research Questions & Hypotheses

The central research question guiding this study is:

RQ1: To what extent can machine learning models predict Instagram post engagement rate using post-level features such as caption length, number of hashtags, posting day and time, and media type?

Engagement rate, calculated as the total number of interactions normalized by reach, provides a standardized measure that reduces bias introduced by differences in audience size. This aligns with best practices in digital marketing analytics and supports valid comparison across posts. The research question is intentionally specific and measurable, consistent with Polonsky & Waller's (2019) guidance that quantitative research questions should clearly connect to the variables being analyzed and the overall purpose of the study.

Primary Hypothesis

H1: Instagram post features have a statistically significant relationship with engagement rate.

Null Hypothesis

H0: Instagram post features have no statistically significant relationship with engagement rate.

Secondary Hypotheses

These secondary hypotheses provide additional analytical depth and support a more granular evaluation of how individual feature groups influence engagement rate outcomes.

H1a: Caption length has a statistically significant relationship with engagement rate.

H1b: Number of hashtags has a statistically significant relationship with engagement rate.

H1c: Posting day/time have a statistically significant relationship with engagement rate.

H1d: Media type (e.g., reel, image, carousel) has a statistically significant relationship with engagement rate.

Together, the research question and hypotheses provide a structured foundation for the analytical approach used in this study and directly support the objectives of developing a predictive model and identifying key engagement drivers.

Literature Review

Introduction to the Literature

The purpose of this study is to examine how Instagram post-level features, such as caption length, hashtags, media type, and posting time predict engagement rate using quantitative modeling. A strong literature review is essential for situating this research within existing scholarship, identifying methodological and conceptual gaps, and establishing the rationale for the study's analytical approach. As Polonsky & Waller (2019) emphasize, a well-constructed literature review provides the theoretical and empirical foundation that guides research design decisions. The following review synthesizes scholarly work on predictive analytics, social media engagement, data quality, and methodological considerations to support the direction of this study.

Predictive Analytics in Social Media Engagement Research

Predictive analytics has become a central tool for understanding user behavior on digital platforms. Shafira et al. (2023) demonstrate that machine learning and regression-based approaches are particularly effective for analyzing large, complex datasets generated by online platforms. Their findings align with Cozzoli et al. (2022), who argue that big data analytics enables organizations to transform heterogeneous and unstructured information into actionable insights. Although Cozzoli et al. focus on healthcare management, the underlying principles of forecasting, pattern recognition, and data-driven decision-making apply broadly to digital engagement research.

Recent studies extend these ideas specifically to social media contexts. Kolhe (2025) found that predictive models using post-level features can accurately estimate Instagram engagement, reinforcing the relevance of feature-based modeling. Similarly, Ju (2024) highlights the growing role of competitive intelligence and algorithmic analysis in shaping social media strategy. Together, these studies support the methodological choice to use regression and predictive modeling to examine how Instagram post characteristics influence engagement outcomes.

Feature-Level Determinants of Instagram Engagement

A consistent theme across the literature is that specific content features significantly influence engagement on social media platforms. Research in digital marketing shows that variables such as media type, caption content, and posting time shape user interactions (Sanches & Ramos, 2025). These findings align with Shafira et al. (2023), who emphasize that engagement patterns are rarely random; instead, they reflect measurable, identifiable features.

Industry practices reinforce these academic insights. Meta platforms use granular post-level features, including timing, hashtags, visual composition, and early interaction patterns, to power recommendation algorithms and optimize content delivery. This alignment between academic research and industry application underscores the applied value of examining how measurable post characteristics predict engagement outcomes.

Data Quality and Preprocessing in Social Media Analytics

Across the literature, scholars emphasize the importance of data quality in predictive analytics. Gray et al. (2021) note that issues such as missing values, outliers, and inconsistent coding can undermine the validity of quantitative research if not addressed systematically. Cozzoli et al. (2022) similarly highlight that big data environments often involve fragmented and unstructured information, requiring robust data cleaning and integration processes.

These insights are particularly relevant for social media data, which frequently contains irregularities due to user-generated content and platform-specific formatting. Studies such as Kolhe (2025) and Sanches and Ramos (2025) also emphasize the importance of preprocessing when modeling engagement, noting that variable standardization and feature engineering directly affect model performance. Collectively, these findings reinforce the need for transparent and rigorous preprocessing procedures in the present study.

Methodological Approaches to Modeling Engagement

Researchers operationalize engagement in different ways, ranging from simple interaction counts to standardized metrics such as engagement rate. These differences influence the comparability of findings and highlight the need for clearly defined metrics in predictive modeling. Methodological approaches also vary widely. Some studies rely on traditional regression models to estimate linear relationships, while others employ machine learning techniques such as random forests or neural networks to capture nonlinear patterns (Shafira et al., 2023).

Comparing these approaches reveals a trade-off between interpretability and predictive power. Regression models offer transparency and ease of interpretation, whereas tree-based models can capture complex interactions but may be less intuitive. This methodological variation underscores the importance of selecting an approach aligned with both the structure of the data and the predictive goals of the study.

Identified Gaps in Existing Research

Although the literature provides strong support for predictive modeling in digital contexts, several gaps remain. First, relatively few studies focus specifically on Instagram engagement rate at the post level; many analyses examine broader platform trends or user-level behavior. Second, methodological rigor varies across studies. While some researchers provide detailed descriptions of preprocessing steps, others omit critical information about how missing data, outliers, or inconsistent metadata were handled. As Gray et al. (2021) and Cozzoli et al. (2022) emphasize, these decisions directly affect internal validity and model performance. Third, studies differ in how they operationalize engagement, making cross-study comparisons challenging. These gaps highlight the need for research that clearly documents data preparation procedures, uses consistent engagement metrics, and evaluates the predictive value of specific post-level features.

While many studies support the use of predictive analytics in social media research, they differ in both methodological approach and the operationalization of engagement. For example, Kolhe (2025) relies on regression-based modeling to estimate engagement using linear relationships, whereas Shafira et al. (2023) demonstrates that machine learning models outperform traditional regression when nonlinear patterns are present. These contrasting findings highlight a methodological divide: regression offers interpretability, while machine learning provides stronger predictive power. Similarly, studies vary in how they define engagement. Some use raw interaction counts, while others, such as Sanches and Ramos (2025), emphasize standardized metrics like engagement rate to account for differences in audience size. These differences underscore the need for consistent metrics and transparent methodological reporting, reinforcing the rationale for the present study's dual-model approach and standardized engagement rate calculation.

Synthesis and Implications for the Present Study

The literature collectively supports the significance and feasibility of examining Instagram post-level features as predictors of engagement. Research on predictive analytics (Shafira et al., 2023), big data methodologies (Cozzoli et al., 2022), data quality (Gray et al., 2021), and social media engagement modeling (Kolhe, 2025) provides a strong foundation for the study's hypotheses and analytical approach. By synthesizing these insights and addressing gaps in methodological transparency and feature-level analysis, this literature review establishes the conceptual and methodological grounding necessary to investigate how measurable post characteristics influence engagement outcomes on Instagram.

Research Design

The research design for this study follows a quantitative, predictive framework intended to examine how Instagram post-level features influence engagement rate and to determine the extent to which these outcomes can be forecasted using statistical and machine learning models. This design is appropriate for the study's objective, which centers on identifying measurable relationships between content characteristics and engagement outcomes while also evaluating the predictive performance of different analytical techniques. The structure of the design emphasizes methodological rigor, transparency, and replicability, ensuring that each stage of the research process, from data preparation to model evaluation, aligns with established quantitative research standards.

This study adopts a post-positivist orientation, assuming that engagement behavior on social media can be partially explained through observable, quantifiable features. The design therefore prioritizes measurable variables, systematic data processing, and objective model evaluation. By integrating both traditional statistical methods and contemporary machine learning approaches, the research design supports a comprehensive examination of engagement dynamics. This dual-model strategy allows the study to balance interpretability with predictive accuracy, reflecting current best practices in digital analytics research.

The design also incorporates a cross-sectional structure, as all observations are drawn from a single dataset representing a snapshot of Instagram activity. This approach is suitable for identifying feature-level patterns and testing hypotheses about the relationships between post characteristics and engagement rate. Although cross-sectional designs do not support causal inference, they are widely used in predictive analytics and social media research due to their efficiency and ability to reveal meaningful associations within large datasets. Overall, the research design provides a coherent and methodologically sound foundation for investigating engagement prediction on Instagram.

Methodology

The methodology outlines the specific procedures used to implement the research design, including data sourcing, variable operationalization, preprocessing, model development, and evaluation. The study uses a publicly available Instagram Analytics Dataset containing post-level observations such as caption length, number of hashtags, media type, posting day and time, impressions, reach, and interaction counts. Engagement rate serves as the dependent variable and is calculated by dividing total interactions by reach, producing a standardized measure that accounts for differences in audience size. This operationalization aligns with established digital marketing practices and supports valid comparison across posts.

Independent variables include caption length, hashtag count, posting time, and media type. Caption length and hashtag count are treated as continuous variables, while posting time and media type are converted into categorical variables to capture temporal and content-based distinctions. Additional features, such as impressions or saves, may be incorporated if supported by model diagnostics and theoretical relevance. These operational definitions ensure consistency and support the assumptions of the analytical techniques used in the study.

The dataset undergoes a rigorous preprocessing workflow to ensure analytical integrity. Missing values related to engagement or reach are removed to maintain accuracy in rate calculations. Outliers in reach or interaction counts are examined and addressed through removal or normalization, depending on diagnostic results. Feature engineering converts posting time into meaningful temporal categories and encodes media type using appropriate categorical techniques. Continuous variables are standardized when required to support model performance. These steps reflect established recommendations in quantitative research and ensure that the dataset is suitable for both regression-based and machine learning models.

The analytical approach integrates multiple modeling techniques to evaluate the predictive value of post-level features. Multiple linear regression is used to assess the statistical significance and direction of relationships between features and engagement rate, offering interpretability and alignment with traditional quantitative methods. Decision tree models are incorporated to capture hierarchical and nonlinear relationships that may not be detected through regression alone. Random forest models further extend this analysis by aggregating multiple decision trees to improve predictive accuracy and reduce overfitting. Feature importance scores derived from the random forest model provide additional insight into the relative influence of each predictor.

Model performance is evaluated using established metrics, including the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE). These measures assess both explanatory power and predictive accuracy. Cross-validation techniques are applied to enhance generalizability and reduce the risk of overfitting, ensuring that model performance is not dependent on a single training subset. Ethical considerations are addressed through the use of publicly available, aggregated post-level data that contain no personally identifiable information. The study adheres to ethical guidelines for responsible data use, transparency, and reproducibility.

Together, these methodological procedures provide a comprehensive and rigorous framework for examining how Instagram post-level features influence engagement rate. By integrating regression and machine learning models within a structured analytical process, the study evaluates both interpretability and predictive performance, supporting the broader goal of generating actionable insights for digital content strategy.

Limitations

Although this study employs a rigorous analytical approach, several limitations must be acknowledged to contextualize the findings and guide interpretation. First, the research relies on a cross-sectional dataset that captures Instagram activity at a single point in time. While this structure is appropriate for identifying feature-level patterns and developing predictive models, it does not allow for causal inference or the examination of temporal changes in engagement behavior.

A second limitation concerns the scope and structure of the dataset itself. The analysis is based on publicly available post-level data, which may not fully represent the diversity of content creators, industries, or audience demographics on Instagram. The dataset does not include user-level characteristics such as follower quality or audience segmentation, which may influence engagement outcomes.

Data quality presents an additional limitation. Although this study incorporates systematic preprocessing procedures, social media data often contain irregularities such as missing metadata or inconsistent formatting. While these issues were addressed through cleaning and feature engineering, it is possible that residual inconsistencies remain. Furthermore, the calculation of engagement rate depends on the accuracy of reach data, which may be influenced by platform-specific reporting practices that are not fully transparent to external researchers.

Modeling constraints also introduce limitations. Regression models assume linearity and may not fully capture complex interactions among features. Tree-based models can identify nonlinear relationships, but may be sensitive to sample size and variable distribution. Although cross-validation was used to mitigate overfitting, generalizability may still be limited.

Finally, the study focuses exclusively on measurable, observable post-level features and does not incorporate qualitative aspects of content such as visual aesthetics or sentiment. These elements may meaningfully influence engagement but require analytical techniques outside the scope of this study. As a result, the findings should be interpreted as reflecting the predictive value of the selected quantitative features rather than the full spectrum of factors that shape engagement on Instagram.

Despite these limitations, the study provides a methodologically sound and analytically transparent examination of how post-level features contribute to engagement outcomes. Acknowledging these constraints strengthens the credibility of the research and highlights opportunities for future studies to further build on this work.

Ethical Considerations

Although the dataset used in this study is publicly available and already anonymized, it is important to consider the ethical implications that would arise if the data had been collected directly from Instagram users. Social media analytics research involves inherent privacy risks because user-generated content can reveal behavioral patterns even when explicit identifiers are removed. If this dataset had been collected firsthand, informed consent would be a central ethical requirement. Users would need to be notified about the purpose of the research, the types of data being collected, and how their information would be stored, analyzed, and protected. Transparency in data collection is essential to ensure that individuals understand how their content contributes to the study and to uphold principles of autonomy and respect for persons.

Another ethical consideration involves the potential for re-identification. Even when usernames or profile information are removed, combinations of post characteristics could theoretically allow a motivated actor to infer identity. Responsible data handling would therefore require safeguards such as data minimization, secure storage, and the removal of metadata that increases re-identification risk. These practices align with broader ethical guidelines for digital research, which emphasize protecting individuals from unintended exposure or harm.

Algorithmic bias also represents an important ethical consideration. Engagement patterns vary widely across creators and industries, meaning predictive models may inadvertently reinforce inequities if the dataset is unbalanced. Ethical research requires acknowledging these limitations and avoiding interpretations that overgeneralize findings.

Finally, the use of predictive insights raises questions about how the results may be applied. While engagement prediction can support content optimization, it can also be misused to encourage manipulative posting strategies. Responsible dissemination requires framing the results as tools for understanding engagement patterns rather than prescriptive rules. Emphasizing transparency, fairness, and respect for user autonomy ensures that the study's outcomes align with ethical standards for digital research.

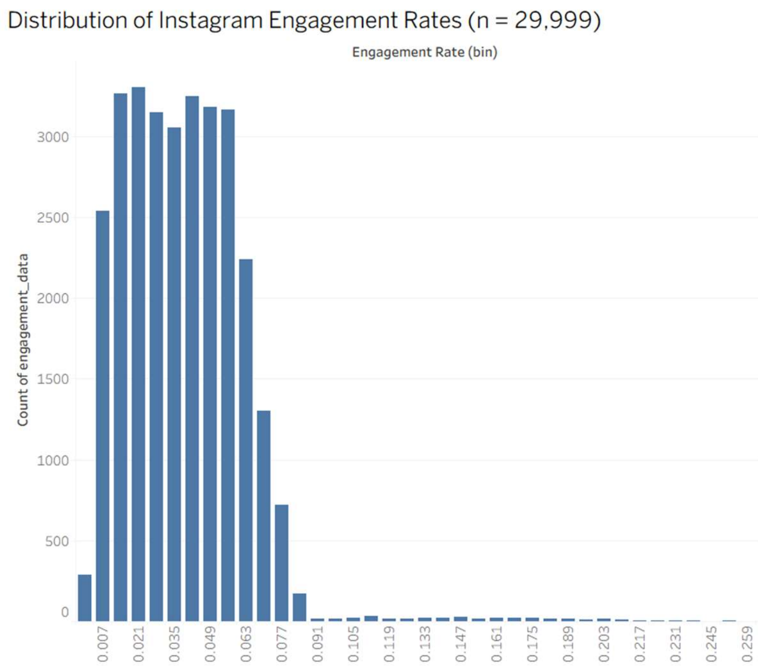
Findings

The purpose of this analysis was to evaluate whether post characteristics including caption length, hashtag count, posting hour, day of week, media type, and content category predict Instagram engagement rate. A combination of descriptive statistics, visual exploration, and regression modeling was used to assess patterns within the dataset. Across all analytical approaches, the results consistently demonstrated minimal variability in engagement rate and no meaningful predictive relationships between the examined features and engagement outcomes.

The analysis began with an examination of the distribution of engagement rate to establish baseline variability. The histogram revealed a narrow distribution centered around approximately 0.04, with limited dispersion and few extreme values. This low variance is analytically important because it constrains the model's ability to detect meaningful relationships; when the dependent variable has little variation, predictors will naturally appear weak.

Figure 1

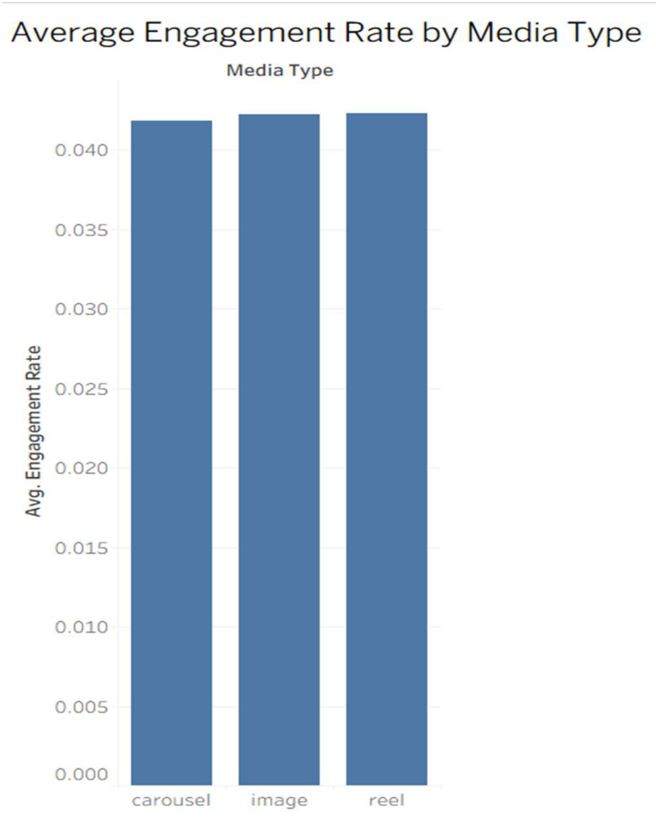
Histogram of Engagement Rate (Tableau)



To explore potential categorical differences, engagement rate was compared across media types. The bar chart showed nearly identical averages for carousel posts, image, and reels, indicating that media format did not influence engagement outcomes.

Figure 2

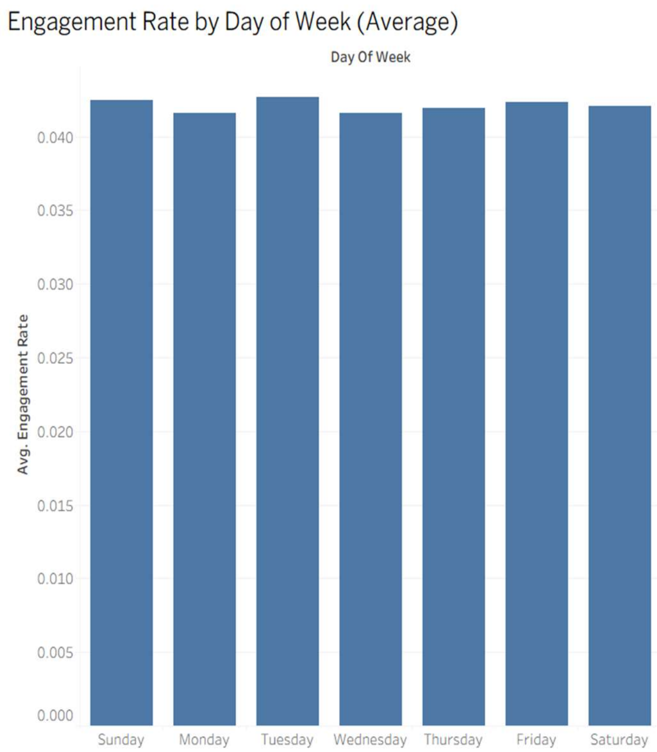
Average Engagement Rate by Media Type (Tableau)



A similar pattern emerged when engagement rate was examined by day of week. All seven days exhibited nearly identical average engagement rates, suggesting that posting day did not meaningfully affect audience interaction.

Figure 3

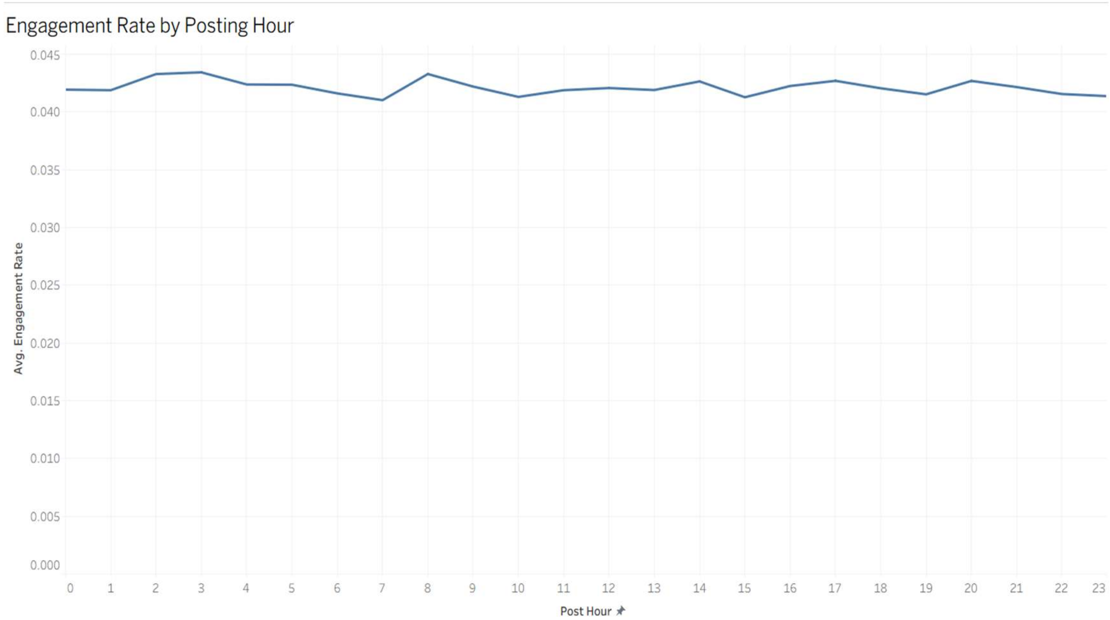
Average Engagement Rate by Day of Week (Tableau)



Temporal patterns were further examined by analyzing engagement rate across posting hours. The line chart revealed small, irregular fluctuations across the 24-hour cycle, but no discernible trend. This stability reinforces the conclusion that posting time did not significantly influence engagement rate.

Figure 4

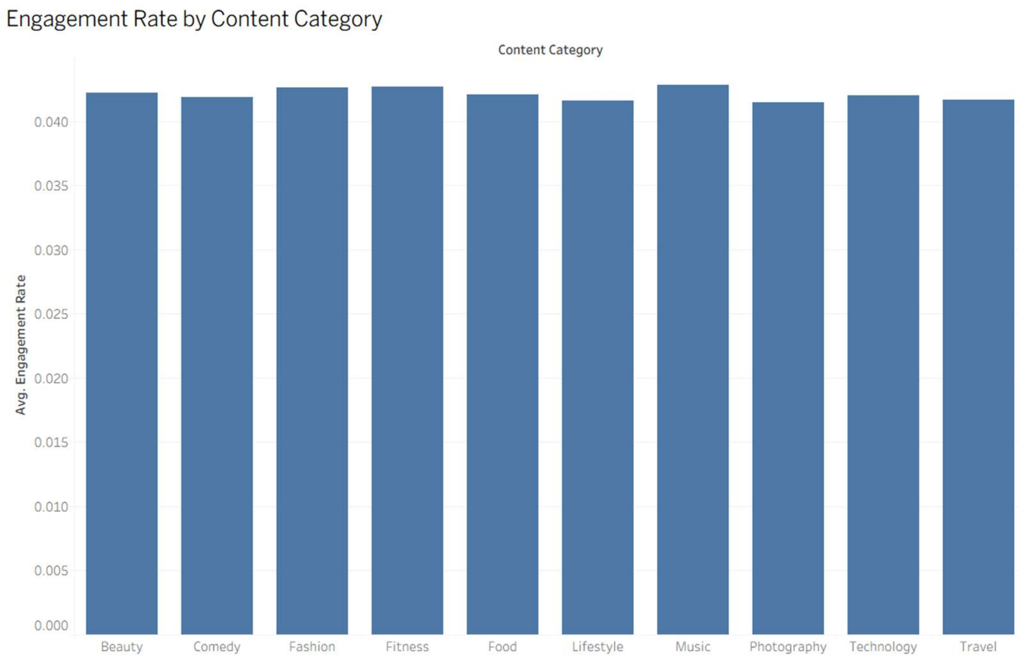
Engagement Rate by Posting Hour



Content category was also evaluated to determine whether certain themes or topics performed better than others. Categories such as Beauty, Comedy, Fashion, Fitness, Food, Lifestyle, Music, Photography, Technology, and Travel all produced nearly identical average engagement rates, which further supports the broader pattern of uniformity in engagement rate outcomes across content types.

Figure 5

Engagement Rate by Content Category



To assess continuous predictors, scatterplots were used to examine the relationships between caption length and engagement rate, and between hashtag count and engagement rate. In both cases, the scatterplots displayed diffuse clouds of points with nearly flat linear trend lines. These visuals indicate that neither the longer captions nor increased hashtag usage systematically influenced engagement rate. The absence of visible patterns in these plots aligns with the regression model's weak, non-significant coefficients for both variables.

Figure 6

Caption Length vs Engagement Rate (Tableau)

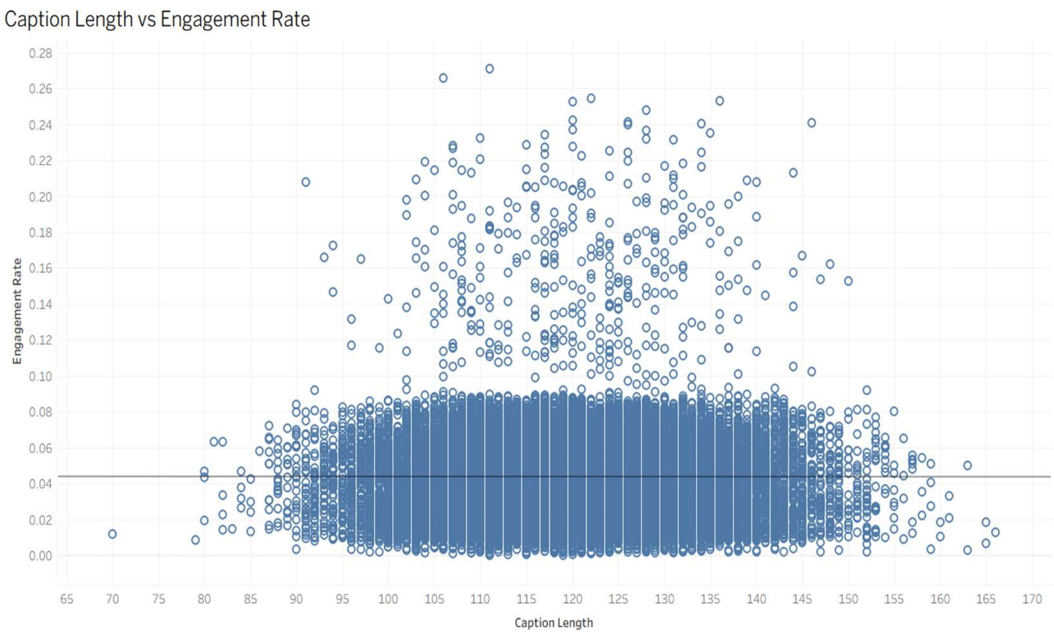
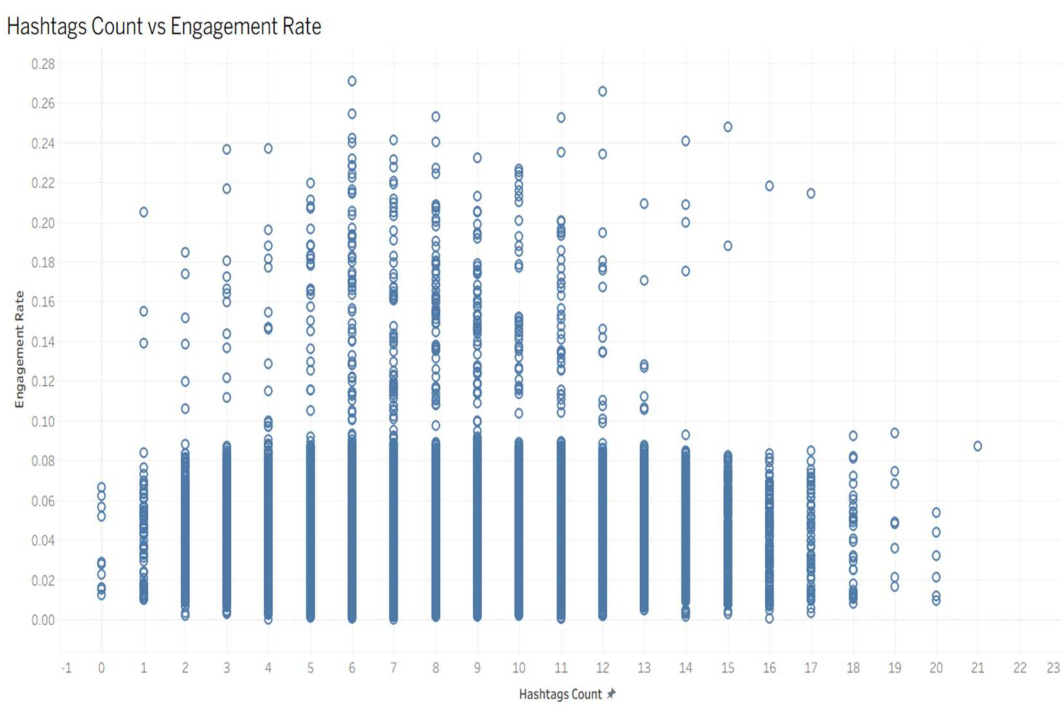


Figure 7

Hashtag Count vs Engagement Rate (Tableau)



The regression model results further confirmed the findings from the descriptive and visual analyses. The multiple linear regression model, estimated in SAS, demonstrated extremely limited explanatory power. The R^2 and adjusted R^2 values were low, indicating that the predictors collectively explained very little variance in engagement rate. The overall F-test was non-significant, suggesting that the model did not perform better than a null model with no predictors.

Figure 8

Model Fit Statistics (SAS)

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	42	0.02399	0.00057124	0.98	0.5005
Error	29956	17.39208	0.00058059		
Corrected Total	29998	17.41607			

Root MSE	0.02410
Dependent Mean	0.04211
R-Square	0.0014
Adj R-Sq	-.0000
AIC	-193493
AICC	-193493
SBC	-223136

Examination of the parameter estimates revealed that nearly all coefficients were small in magnitude and statistically non-significant, with p-values well above conventional thresholds. These results are consistent with the earlier visual findings, which showed minimal differences across categories and no discernible relationships with continuous predictors.

Figure 9

Parameter Estimates Table (SAS)

Parameter Estimates

Parameter	DF	Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	0.040454	0.001819	22.24	<.0001
caption_length	1	0.000001951	0.000012645	0.15	0.8774
hashtags_count	1	-0.000012300	0.000049227	-0.25	0.8027
post_hour 0	1	0.000537	0.000966	0.56	0.5785
post_hour 1	1	0.000517	0.000975	0.53	0.5957
post_hour 2	1	0.001903	0.000982	1.94	0.0526
post_hour 3	1	0.002058	0.000970	2.12	0.0339
post_hour 4	1	0.001028	0.000968	1.06	0.2879
post_hour 5	1	0.001010	0.000978	1.03	0.3017
post_hour 6	1	0.000239	0.000976	0.24	0.8067
post_hour 7	1	-0.000362	0.000967	-0.37	0.7081
post_hour 8	1	0.001917	0.000973	1.97	0.0487
post_hour 9	1	0.000861	0.000971	0.89	0.3755
post_hour 10	1	-0.000061875	0.000967	-0.06	0.9490
post_hour 11	1	0.000524	0.000965	0.54	0.5875
post_hour 12	1	0.000688	0.000969	0.71	0.4774
post_hour 13	1	0.000499	0.000966	0.52	0.6054
post_hour 14	1	0.001246	0.000965	1.29	0.1968
post_hour 15	1	-0.000072483	0.000966	-0.08	0.9402
post_hour 16	1	0.000873	0.000979	0.89	0.3726
post_hour 17	1	0.001347	0.000973	1.38	0.1662
post_hour 18	1	0.000662	0.000974	0.68	0.4967
post_hour 19	1	0.000182	0.000970	0.19	0.8511
post_hour 20	1	0.001338	0.000967	1.38	0.1665
post_hour 21	1	0.000790	0.000981	0.81	0.4208
post_hour 22	1	0.000154	0.000966	0.16	0.8731
post_hour 23	0	0	.	.	.
day_of_week Friday	1	0.000761	0.000521	1.46	0.1443
day_of_week Monday	1	-0.000001490	0.000519	-0.00	0.9977
day_of_week Saturday	1	0.000516	0.000520	0.99	0.3215
day_of_week Sunday	1	0.000849	0.000522	1.63	0.1035
day_of_week Thursday	1	0.000327	0.000517	0.63	0.5269
day_of_week Tuesday	1	0.001046	0.000517	2.02	0.0431
day_of_week Wednesday	0	0	.	.	.
media_type carousel	1	-0.000435	0.000364	-1.19	0.2327
media_type image	1	-0.000012030	0.000356	-0.03	0.9731
media_type reel	0	0	.	.	.
content_category Beauty	1	0.000518	0.000627	0.83	0.4084
content_category Comedy	1	0.000191	0.000627	0.30	0.7604
content_category Fashion	1	0.000927	0.000622	1.49	0.1362
content_category Fitness	1	0.001041	0.000624	1.67	0.0953
content_category Food	1	0.000423	0.000624	0.68	0.4973
content_category Lifestyle	1	-0.000036062	0.000623	-0.06	0.9539
content_category Music	1	0.001146	0.000624	1.84	0.0663
content_category Photography	1	-0.000203	0.000622	-0.33	0.7437
content_category Technology	1	0.000360	0.000623	0.58	0.5639
content_category Travel	0	0	.	.	.

Although the model met the structural requirements for linear regression, the results collectively reinforce the central finding: the dependent variable exhibited very low variability, limiting the model's ability to detect meaningful relationships. Taken together, the descriptive statistics, visual analyses, and regression modeling converge on the same conclusion: none of the examined post characteristics meaningfully predicted engagement rate in this dataset. Engagement outcomes remained stable regardless of media type, posting day, posting hour, content category, caption length, or hashtag count. These findings suggest that engagement rate may be driven by factors not captured in the dataset, such as creator influence, audience characteristics, content quality, or platform-level algorithmic dynamics. The regression results directly address Research Question 1 by demonstrating that the selected post-level features do not provide meaningful predictive value for engagement rate. The absence of statistically significant relationships indicates that the examined variables cannot reliably forecast engagement outcomes within this dataset.

Conclusion

This study examined the extent to which measurable Instagram post characteristics could explain or predict engagement rate. After conducting descriptive analysis, visual exploration, and regression modeling, the results demonstrated a consistent pattern: engagement rate remained highly uniform across all examined features. The narrow distribution of engagement rate, centered around approximately 0.04, limited the potential for strong predictive relationships and shaped the overall outcome of the analysis.

Across categorical variables including media type, posting day, posting hour, and content category, engagement levels showed minimal variation, indicating that these factors did not meaningfully influence audience interaction. Continuous predictors such as caption length and hashtag count similarly showed no discernible relationship with engagement, as evidenced by flat trend lines and diffuse scatterplot patterns. These visual findings aligned with the regression model, which produced low explanatory power, non-significant coefficients, and no evidence that the included predictors contributed meaningfully to engagement rate outcomes.

Taken together, the results indicate that the variables available in this dataset do not meaningfully predict engagement rate. Instead, engagement appears to be shaped by factors not captured in the current data, such as creator influence, audience characteristics, content quality, or platform-level algorithmic dynamics.

While the model met the structural requirements for linear regression, the limited variability in the engagement rate constrained the ability to detect meaningful relationships. Based on the regression results and the absence of meaningful relationships across all examined variables, the null hypothesis, that Instagram post characteristics do not significantly predict engagement rate, cannot be rejected. These findings underscore the importance of incorporating additional or alternative predictors in future analyses to better understand the drivers of engagement on social media platforms.

Recommendations

Building on these findings, several practical and methodological recommendations emerge for practitioners and future researchers seeking to better understand engagement dynamics. The results of this analysis indicate that the Instagram post characteristics included in the dataset did not meaningfully predict engagement rate. Based on these findings, several recommendations can be made for practitioners and future researchers seeking to better understand or improve engagement outcomes.

First, content creators and social media managers should avoid relying solely on structural post characteristics (such as posting time, caption length, or hashtag volume) as primary strategies for increasing engagement. The stability of engagement across all examined features suggests that these variables, while commonly emphasized in social media best-practice guides, may not meaningfully influence audience behavior in isolation. Instead, practitioners should prioritize factors not captured in this dataset but widely recognized in industry research as influential, such as content quality, storytelling clarity, visual appeal, and alignment with audience interests. These elements are more likely to shape user engagement than mechanical posting attributes.

Second, brands should consider incorporating audience-specific insights into their content strategies. Engagement is often driven by the relationship between the creator and their audience, including trust, familiarity, and perceived authenticity. Metrics such as follower demographics, audience growth patterns, and historical engagement trends may offer more actionable insights than the structural variables analyzed in this study. Integrating these audience-level indicators into future analyses may provide a more nuanced understanding of what drives engagement.

Third, future research should expand the scope of available predictors to include variables that were not present in the current dataset but are likely to influence engagement. Examples include sentiment analysis of captions, visual content features (e.g., brightness, color palette, facial presence), creator-level metrics (e.g., follower count, posting frequency), and algorithmic factors such as reach or impressions. Incorporating these additional predictors may improve model performance and provide a more comprehensive understanding of engagement dynamics.

Finally, researchers should consider using alternative modeling approaches that may be better suited to low-variance outcomes. An alternative modeling technique, such as gradient boosting, may capture non-linear relationships that linear regression cannot detect. Additionally, collecting a larger or more diverse dataset may help increase variability in engagement rate, thereby improving the potential for predictive modeling.

Overall, the findings of this study suggest that practitioners should shift their focus away from isolated posting attributes and toward more holistic, audience-centered, and content-quality-driven strategies. Future research that incorporates richer predictors and more advanced modeling techniques may yield deeper insights into the complex factors that shape engagement on social media platforms.

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